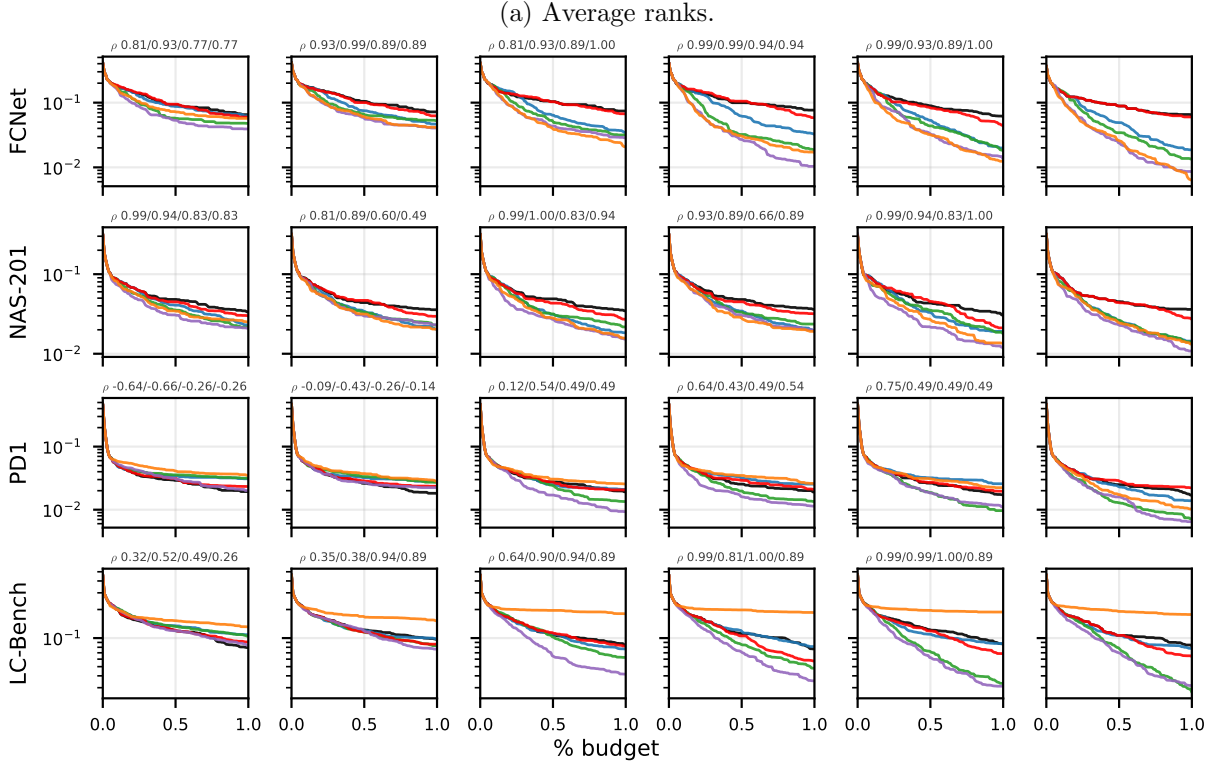
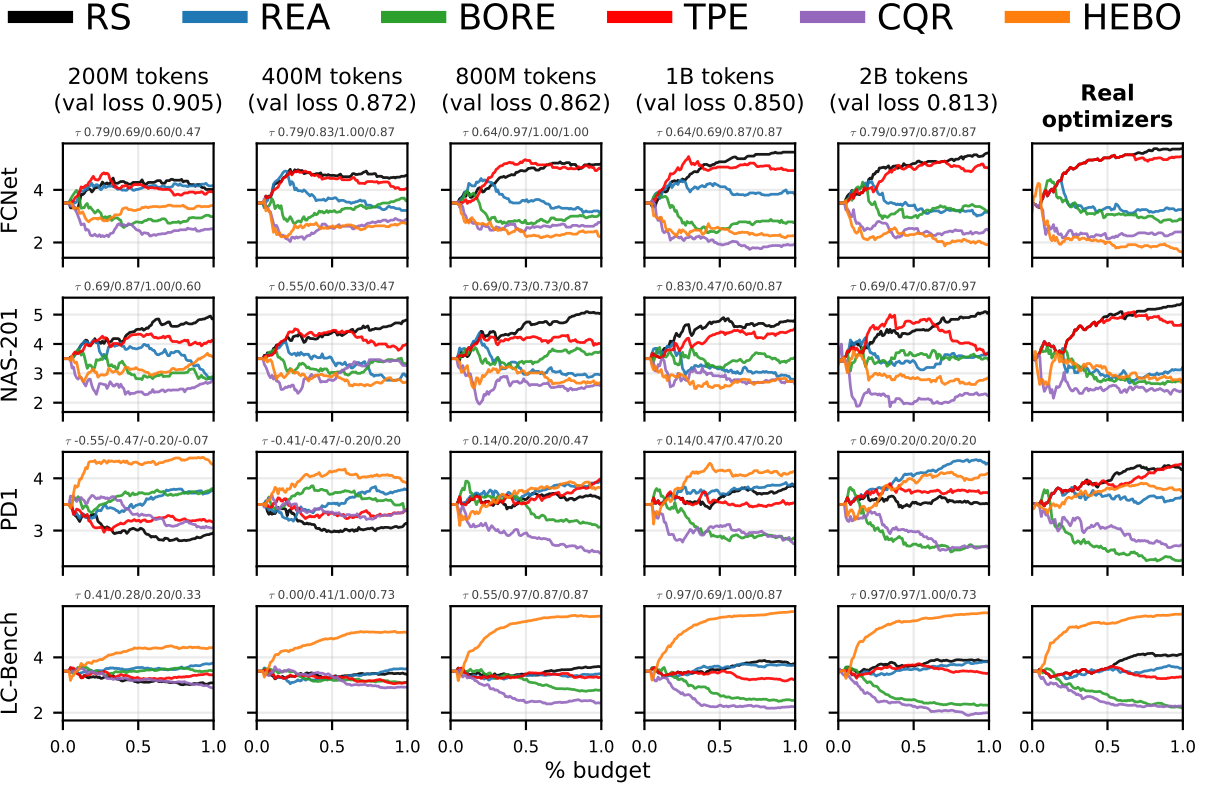




Imitation quality vs. training-token budget for the 80M model. Average rank (a) and normalized regret (b) of the imitated optimizers (columns: 200M/400M/800M/1B/2B training tokens) against the real optimizers (last column), on the same four held-out validation families (30 seeds each). Column headers report validation loss (lower is better), improving from 0.905 (200M tokens) to 0.813 (2B tokens). And, on top of each plot, we highlight the correlation with the real optimizers at 25/50/75/100% budget.